**Adam**

**input :** starting point $\vec{x}_0$, step factor $\alpha$, decay rates $\gamma_v$, $\gamma_s \in [0, 1)$, small constant $\epsilon$, tolerance $E_s$

**output:** $\vec{x}^*$, estimate of the minimizer, and $f(\vec{x}^*)$

// defaults from the original paper: $\alpha = 0.001$, $\gamma_v = 0.9$, $\gamma_s = 0.999$, $\epsilon = 10^{-8}$

// all vector products, square roots, and divisions below are elementwise

$\vec{v}_0 \leftarrow \vec{0}$, $\vec{s}_0 \leftarrow \vec{0}$, $k \leftarrow 0$

**while** $\|\nabla f(\vec{x}_k)\|_\infty > E_s$ **do**

$\vec{g}_k \leftarrow \nabla f(\vec{x}_k)$

$\vec{v}_{k+1} \leftarrow \gamma_v \vec{v}_k + (1 - \gamma_v) \vec{g}_k$ // biased decaying momentum

$\vec{s}_{k+1} \leftarrow \gamma_s \vec{s}_k + (1 - \gamma_s) (\vec{g}_k \odot \vec{g}_k)$ // biased decaying sq. gradient

$\hat{\vec{v}}_{k+1} \leftarrow \vec{v}_{k+1} / (1 - \gamma_v^{k+1})$ // corrected decaying momentum

$\hat{\vec{s}}_{k+1} \leftarrow \vec{s}_{k+1} / (1 - \gamma_s^{k+1})$ // corrected decaying sq. gradient

$\vec{x}_{k+1} \leftarrow \vec{x}_k - \alpha \hat{\vec{v}}_{k+1} / \left(\epsilon + \sqrt{\hat{\vec{s}}_{k+1}}\right)$ // next iterate

$k \leftarrow k + 1$

**end**

**return** $\vec{x}^* = \vec{x}_k$, $f(\vec{x}^*)$